MIL-G-13451B(EL)
23 SEPTEMBER 1966
SUPERSEDING
MIL-G-13451A(SigC)
19 AUGUST 1954

MILITARY SPECIFICATION

GAGES, DIMENSIONAL CONTROL, MECHANICAL (FOR USAECOM)

1. SCOPE

1.1 This specification covers procurement requirements for gages and gage cases.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications of the issue in effect on date of invitations for bids, form a part of this specification.

SPECIFICATIONS

MILITARY

MIL-G-10944 — Gages, Dimensional Control

(Copies of Military Specifications or Ordnance Standards may be obtained from, or as directed by, the procuring agency. Both title and identifying symbol or number should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 General. Unless otherwise specified in the bid request and contract, and except as specified below, the applicable requirements of Specification MIL-G-10944 shall apply for all gages fabricated in accordance with this specification.

3.1.1 The limits of the part to be gaged, as specified on the drawings, shall not be exceeded as a result, either of tolerance or wear of the gages. Therefore, the extreme sizes for all gages shall not exceed the extreme limits of the part to be gaged. All variations in the gage, whatever their cause or purpose, shall bring the gage within these extreme limits. Thus a gage that checks a minimum dimension may be larger, but never smaller, than the minimum size specified for the part to be gaged. Likewise, the gage that checks the maximum dimensions may be smaller, but never larger, than the maximum size specified for the part to be gaged.

3.1.2 Any part which is so close to either rejection limit as to be improperly rejected, either as a result of tolerance or wear of the inspection gages, can be reinspected by the contractor using a gage as close to the product limit as is practicable, or by use of precision measuring instruments and shown to the Government inspector. Any observational errors in the reinspection must, however, be in the direction of safety rather than in the direction of danger of acceptance of improper parts.

3.1.3 *Hardness.* Where the gage contact or wearing surface hardness is not specified in procurement documents, the following hardness requirements shall apply based on the finished section of the part.

Thickness up to .0625 ... Rockwell C50 to C55

FSC 5999

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Thickness .0625 to .125 (inclusive)	Rockwell C55 to C60
Thickness .125 to .1875 (inclusive)	Rockwell C60 to C63
Thickness above .1875	Rockwell C63 to C66
Threads and Serrated gages	Rockwell C60 to C63
Spring steel	Rockwell C45 to C55
Machine steel	Rockwell 15N 90 MIN (Case or pack harden .02 min. depth after grinding.)

3.1.4 *Surface quality.* When the surface quality is not specified in drawings listed in the bid request and contract, the quality of gaging surfaces shall conform to good gage making practices, shall be no rougher than 8 micro-inches, and shall be finished by lapping or polishing.

3.1.5 *Marking (identification data).* Gages covered by MIL Standards shall be marked as specified in the Standards. Except as otherwise specified in gage drawings listed in the bid request and contract, the following information shall be shown on the gages furnished. (See figure 1 for example of form.)

(a) Nomenclature type number of nearest related nomenclatured component on which gage is to be used.

(b) Name of part to be gaged.

(c) Function of gage with dimension to be checked.

(d) Kind of gage: Sig C (Insp. Master etc.).

(e) Gage number: (To be assigned by the contracting officer).

(f) Manufacturer's name or symbol.

3.1.5.1 *Type of lettering.* Letters shall be Gothic capitals, and the numerals and characteristics shall be of similar appearance. Letters, numerals, and other characteristics shall be of such size as to be clearly legible.

3.1.5.2 Gages shall be mechanically stamped or engraved directly in a permanent manner. Where space does not permit all the information specified above to be marked, the contractor shall confer with the contracting officer to determine the markings to be included.

3.2 Gages and Gage Cases.

3.2.1 Gages and gage cases produced for the order shall be submitted for acceptance at least 45 calendar days prior to scheduled delivery date of first production equipments. The gages shall be accompanied by a copy of the measurements and data, as made by the gage manufacturer, confirming that the gages are in accordance with the applicable drawings. Inspection of these gages and cases will require not more than 30 calendar days. If any gages or cases are disapproved, the rejected gages or cases shall be returned to the contractor. The contractor shall expedite the submission of required gages or cases for reinspection and retest. At the option of the contracting officer, rejected gages or cases shall be reworked, or replaced by the contractor, with new gages or cases. Acceptance of production equipment will be contingent upon the prior delivery of approved specified gages for availability and use by Government inspector. Gage cases shall be submitted for acceptance together with the gages.

3.3 Special devices. All special devices, such as pins, buttons, templates etc., used by the contractor for inspection, adjustment, or calibration of a particular gage, shall be furnished by the contractor, shall

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become the property of the Government, and shall be shipped with the required gages.

3.4 Workmanship. The workmanship shall be first class, consistent with the tolerances and finishes specified, and with the best gage manufacturing practices.

4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection and test, general. Unless otherwise specified in the bid request and contract, the inspection and tests of gages fabricated in accordance with this specification shall be in accordance with Specification MIL–G–10944. Inspections and tests shall be made at the location specified in the bid request and contract. (See 6.4b.)

5. PREPARATION FOR DELIVERY

5.1 Unless otherwise specified in the bid request and contract, the preparation for delivery shall be in accordance with Specification MIL–G–10944.

6. NOTES

6.1 Intended use. The gages covered by this drawing are intended for use by Government inspectors for inspection of USAECOM material.

6.2 The "NOTES" of Specification MIL–G–10944 apply with the following addition to the "Ordering data:"

"f. Date for final acceptance of gages (3.2.1)"

6.3 Gages, as generally referenced in this specification, includes test fixtures.

6.4 Submissions and inquiries.

6.4a All gage and gage case shipments made in connection with this specification should be made to:

Supply Support Services Ofc.
Bldg. 117, Fort Monmouth, N. J. 07703
M/F U. S. Army Electronics Command
Gage Engineering Branch
Nomenclature .....
Order: # .....

6.4b All inquiries, submissions and shipments made in connection with this specification should be made to:

Commanding Officer
U. S. Army Electronics Command
Coles Area
Fort Monmouth, N. J. 07703
ATTN: Gage Engineering Branch, AMSEL–PP–EE–3
Elct. Warfare & Met. Division

Custodian:
Army—GL

Preparing activity:
Army—EL
Project No. 5999–A013

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Example of Form of Markings on Gages
(For USAECOM Equipment)

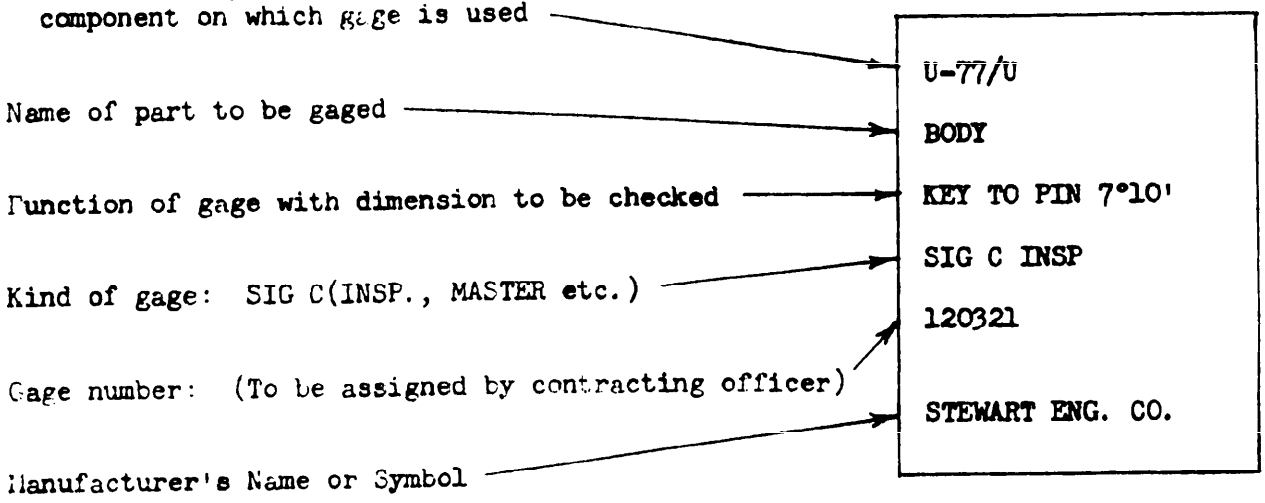


FIGURE 1
(see paragraph 3.1.5)

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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